recent policy results and PDP connections locally, ensuring quick enforcement even under high load. This distributed architecture prevents any request—regardless of its origin—from crossing tenant boundaries or bypassing ACME's governance controls.

13.5.2 Tenant-level enforcement: collaboration services

After a request passes through the global gateway, enforcement continues within the tenant's environment. Each tenant's workspace is hosted across a set of microservices along with document, folder, and workspace APIs that together form the core of C³.

Each service has its own local PEP, either as a sidecar container or a shared SDK, that intercepts calls to protected operations. These tenant PEPs perform three coordinated functions:

1. **Translate application actions**. They map UI or API operations—such as view, edit, or share—to the corresponding Cedar policy actions.
2. **Consult the tenant's PDP**. Each PEP contacts the tenant's PDP (either a shared cluster or a dedicated instance) to evaluate ReBAC policies for that tenant's namespace.
3. **Enforce results locally**. The PEP uses the decision to ensure that business logic executes only when authorized.

Because the PEPs operate alongside or within the services that manage the data, they can cache entity graphs and policy scopes in memory, which reduces round-trips and latency. Each PEP updates its cache asynchronously from the tenant's entity store, ensuring ReBAC decisions stay current without slowing user operations.

Together, the gateway PEPs at the platform edge and the tenant PEPs inside collaboration services create a distributed enforcement mesh. This design enables C³ to evaluate policies near the data they protect, scale horizontally as tenants grow, and ensure consistent semantics across every layer of the platform.